The Confidence Cost of Pairwise Distribution Learning

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Abstract

Pairwise conditional access to an unknown distribution returns a draw from the distribution restricted to any requested pair. At constant confidence, this weak oracle is as sample-efficient for labeled total-variation learning as ordinary sampling, up to constants. We show that this equivalence fails at high confidence.

For a distribution on n labels, accuracy ε, and failure probability δ, we prove the exact minimax query complexity

$$\Theta\left(\frac{n\log(1/\delta)}{\varepsilon^2}\right).$$

The corresponding ordinary-sampling complexity is $\Theta((n+\log(1/\delta))/\varepsilon^2)$. Thus confidence amplification costs up to a factor n under pairwise access.

The upper bound reuses one trajectory of a pairwise-winner Markov chain whose stationary law is the target. Although its worst-case relaxation time is linear in n, we prove that its empirical distribution has the i.i.d.-scale stationary risk $O(\sqrt{n/T})$. The proof compares its conductances with a minimum-mass graph. After sorting the target probabilities, this graph has an explicit Helmert eigensystem, and a capped-mass Green-function bound telescopes without any minimum-mass or dynamic-range assumption. A geometric median of independent trajectories gives the high-confidence result. The lower bound uses a heavy label and uniform light labels. Every adaptive pair query then carries only $O(\varepsilon^2/n)$ KL information, which makes the multiplicative confidence cost unavoidable.

1 Introduction

Suppose an unknown distribution p on $[n]$ cannot be sampled directly. Instead, a learner chooses two labels i, j and observes a draw from p conditioned on $\{i, j\}$. This pairwise conditional, or PCOND, oracle is the sampling form of the Bradley–Terry–Luce choice model (Bradley and Terry, 1952; Luce, 1959). It appears when providers expose only local menus, when comparisons are easier to elicit than calibrated scores, and when global samples cannot be observed.

Pairwise access identifies every labeled distribution, including targets with arbitrarily large dynamic range. The quantitative picture was recently clarified by Kleinberg et al. (2026). Their structural theory of restricted providers gives

$$\Omega\left(\frac{n+\log(1/\delta)}{\varepsilon^2}\right) \le q_{\text{pair}} \le O\left(\frac{n\log^2 n\log(n/\delta)}{\varepsilon^2}\right), \tag{1}$$

and asks whether the remaining polylogarithmic gap can be closed. Existing comparison estimators do not settle this question. Their useful uniform bounds generally impose a bounded parameter range, a fixed passive design, or a different loss on log qualities (Shah et al., 2015; Agarwal et al., 2018; Hendrickx et al., 2019, 2020).

We close the gap, but the answer differs from the ordinary-sampling benchmark. Pairwise learning has a multiplicative, not additive, dependence on confidence.

Theorem 1 (Minimax pairwise learning)**.** *There are universal constants $c, C > 0$ such that, for $n \ge 4$, $0 < \varepsilon \le 1/16$, and $0 < \delta \le 1/8$,*

$$c\frac{n\log(1/\delta)}{\varepsilon^2} \le q_{\text{pair}}(n,\varepsilon,\delta) \le C\frac{n\log(1/\delta)}{\varepsilon^2}. \tag{2}$$

The upper bound is attained by a polynomial-time adaptive learner and allows arbitrary zero probabilities.

For comparison, ordinary i.i.d. sampling has complexity $\Theta((n+\log(1/\delta))/\varepsilon^2)$ (Canonne, 2020). At constant

Table 1: Minimax labeled TV learning rates.

Observation model	query or sample complexity
Ordinary samples	$\Theta((n+\log(1/\delta))/\varepsilon^2)$
Pairwise conditional	$\Theta(n\log(1/\delta)/\varepsilon^2)$
Prior pairwise upper	$O(n\log^2 n\log(n/\delta)/\varepsilon^2)$

confidence the two models both require $\Theta(n/\varepsilon^2)$ observations. If $\log(1/\delta) \gg n$, pairwise access requires a factor $\Theta(n)$ more. Table 1 summarizes the separation.

Why mixing time is misleading. The upper bound starts from a natural chain. From state i, propose a uniformly random $j \neq i$, query $\{i, j\}$, and retain the returned winner. Its stationary law is p. Restarting after each $O(n)$-step burn-in gives a quadratic learner. A single trajectory has worst-case relaxation time $\Theta(n)$, so a black-box spectral-gap bound still suggests quadratic cost. The slow mode, however, occurs only when its stationary mass is small or concentrated in a few coordinates. Total variation automatically weights these modes by their statistical relevance.

Our main technical result makes this compensation exact. We dominate the chain's Green function by that of the complete graph with edge conductance $\min(p_i, p_j)/n$. Once $p_1 \geq \cdots \geq p_m > 0$, this graph is diagonalized by the Helmert contrasts. Its eigenvalues are capped masses

$$c_k/n, \qquad c_k = kp_k + \sum_{j>k} p_j = \sum_j \min\{p_j, p_k\}.$$

The coordinate Green functions then split into left, local, and right terms. Each aggregate term is bounded by one telescoping inequality. This yields $O(\sqrt{n/T})$ TV risk uniformly over the whole simplex.

Why confidence is different. Take one label of mass near $1/2$ and spread the remaining mass uniformly over $n-1$ labels. Changing the heavy mass by $\Theta(\varepsilon)$ changes the outcome probability of every informative pair by only $\Theta(\varepsilon/n)$, while that outcome itself has probability $\Theta(1/n)$. Each query therefore contributes only $O(\varepsilon^2/n)$ KL divergence. This obstruction applies transcript by transcript to adaptive learners and forces the product $n\log(1/\delta)/\varepsilon^2$.

Contributions. Our results are entirely theoretical.

1. We determine the minimax PCOND complexity at every confidence level, closing the dimension and accuracy gap in (1).
2. We prove a distribution-free occupation-risk theorem for the winner chain. The result has no lower mass, bounded ratio, or warm-start assumption.

Algorithm 1 High-confidence pairwise learner

Set $C_0 = 2+2\sqrt{2}$, $B = \lceil (n-1)\log(48/\varepsilon)\rceil$, $T = \lceil (48^2 C_0^2 n/\varepsilon^2\rceil$, and $R = \lceil 18\log(1/\delta)\rceil$.
for $r = 1, \ldots, R$ **do**
  Start at an arbitrary $X_0 \in [n]$.
  for $t = 0, \ldots, B+T-1$ **do**
    Draw J_t uniformly from $[n] \setminus \{X_t\}$.
    Query $\{X_t, J_t\}$ and set X_{t+1} to its winner.
  end for
  Let $\widehat{p}_r$ be the empirical law of $X_{B+1}, \ldots, X_{B+T}$.
end for
return an L_1 geometric median of $\widehat{p}_1, \ldots, \widehat{p}_R$.

3. We identify a confidence separation between pairwise conditional and ordinary sampling, and prove that adaptive pair selection cannot remove it.

2 Model and Algorithm

Let Δ_n be the simplex on $[n]$. For $p \in \Delta_n$, a query to an unordered pair $\{i, j\}$ returns i with probability $p_i/(p_i+p_j)$ and j otherwise. If both masses vanish, the response may follow any fixed convention. A learner may choose each pair using its past queries, responses, and private randomness.

We define $q_{\text{pair}}(n, \varepsilon, \delta)$ as the least fixed query budget for which some learner returns $\widehat{p}$ satisfying

$$\sup_{p\in\Delta_n} \mathbb{P}_p\{d_{\mathrm{TV}}(\widehat{p}, p) > \varepsilon\} \leq \delta,$$

$$d_{\mathrm{TV}}(p, q) = \tfrac{1}{2}\|p-q\|_1.$$

One base estimator follows a single winner-chain trajectory. Repeating it and aggregating by a metric median gives Algorithm 1. An L_1 geometric median is any minimizer over Δ_n of the sum of distances to the candidates. It is computable by a linear program.

The stated constants simplify the proof and are not optimized. Since $\log(48/\varepsilon) = O(1/\varepsilon^2)$, the total query count has the order in Theorem 1.

2.1 What the algorithm changes

The global chain itself is not new. It appears in the generic complete-family learner of Kleinberg et al. (2026). The difference is which states are kept and how they are analyzed. That learner runs the chain for $B = \Theta(n\log(1/\varepsilon))$ steps, keeps only the terminal state, and restarts independently $M = \Theta((n+\log(1/\delta))/\varepsilon^2)$ times. Even in the pairwise case, this costs

$$\Theta\left(\frac{n(n+\log(1/\delta))\log(1/\varepsilon)}{\varepsilon^2}\right)$$

queries. Their faster hierarchical learner keeps trajectories inside many tree cells, then reconstructs p from binary split estimates. It pays two tree-depth logarithms and a simultaneous-confidence logarithm.

Algorithm 1 instead burns in one global chain and retains every later state. This removes repeated mixing, but standard MCMC analysis does not justify the improvement. The uniform minorization in (17) gives gap only $1/(n-1)$. Applying that gap to each coordinate and summing gives the unhelpful bound $O(n/\sqrt{T})$. The next section replaces this worst-mode argument by a joint coordinate calculation.

3 One Trajectory Has i.i.d.-Scale Risk

We first analyze a stationary trajectory. On the positive support of p, the winner chain has transition kernel

$$P(i,j) = \frac{1}{n-1}\frac{p_j}{p_i+p_j}, \quad i \neq j, \tag{3}$$

with the diagonal chosen to make rows sum to one. Detailed balance holds because

$$p_i P(i,j) = \frac{p_i p_j}{(n-1)(p_i+p_j)} = p_j P(j,i). \tag{4}$$

Theorem 2 (Stationary occupation risk)**.** *Let $(X_t)_{t=1}^T$ be a stationary trajectory of (3), and let $\widehat{p}_T$ be its empirical law. For every $p \in \Delta_n$,*

$$\mathbb{E}_p d_{\mathrm{TV}}(\widehat{p}_T, p) \leq (2+2\sqrt{2})\sqrt{\frac{n}{T}}. \tag{5}$$

The result is not a consequence of the worst-case gap, which can be $\Theta(1/n)$. We prove it through coordinate-specific Green functions. Let $f_i(x) = \mathbf{1}\{x = i\} - p_i$ and

$$H_i = \langle f_i, (I-P)^{-1} f_i \rangle_p, \tag{6}$$

where the inverse acts on mean-zero functions in $L_2(p)$.

There is a useful electrical interpretation. Let $D = \mathrm{diag}(p)$ and let $L = D(I-P)$ be the weighted graph Laplacian associated with (4). Since

$$D f_i = p_i(e_i - p) =: b_i,$$

the coordinate Green function is

$$H_i = b_i^\top L^\dagger b_i. \tag{7}$$

Thus H_i is the minimum energy needed to send source $p_i(1-p_i)$ from i and absorb $p_i p_j$ at every $j \neq i$. A small global gap can coexist with small coordinate energy because each source is scaled by its target mass.

Lemma 3 (Positive spectrum and variance)**.** *The kernel P is positive semidefinite on $L_2(p)$. Moreover,*

$$\mathrm{Var}_p\left(\frac{1}{T}\sum_{t=1}^T f_i(X_t)\right) \leq \frac{2H_i}{T}. \tag{8}$$

For positive semidefiniteness, use

$$\frac{p_i p_j}{p_i+p_j}(f_i - f_j)^2 \leq p_i f_i^2 + p_j f_j^2$$

and sum over pairs. Reversibility then gives eigenvalues in $[0,1]$. Expanding an additive functional in the eigenbasis and summing its geometric covariances proves (8).

It remains to control all H_i jointly. Let $m = |\operatorname{supp}(p)|$ and relabel the positive masses so that $p_1 \geq \cdots \geq p_m > 0$. Define

$$c_k = k p_k + \sum_{j>k} p_j. \tag{9}$$

Lemma 4 (Capped-mass Green bound)**.** *For $i \in [m]$, define*

$$A_i = \sum_{k=2}^{i-1} \frac{(1-c_k)^2}{k(k-1)c_k} + \mathbf{1}\{i \geq 2\}\frac{(i-c_i)^2}{i(i-1)c_i} + \sum_{k=i+1}^{m} \frac{c_k}{k(k-1)}. \tag{10}$$

Then

$$H_i \leq 2n p_i^2 A_i, \qquad \sum_{i=1}^m p_i\sqrt{A_i} \leq 2 + 2\sqrt{2}. \tag{11}$$

Proof idea. The Dirichlet conductance in (4) obeys

$$\frac{p_i p_j}{(n-1)(p_i+p_j)} \geq \frac{\min\{p_i,p_j\}}{2n}. \tag{12}$$

Let L_0 be the graph Laplacian with conductance $\min\{p_i,p_j\}/n$. The Helmert vectors

$$u_k = \frac{(\underbrace{1,\ldots,1}_{k-1}, -(k-1), 0, \ldots, 0)}{\sqrt{k(k-1)}} \tag{13}$$

diagonalize L_0, with $L_0 u_k = (c_k/n) u_k$. Projecting the coordinate source $p_i(e_i - p)$ onto this basis gives (10) exactly. The factor two in (11) follows from (12).

For completeness, the projections are explicit. One has

$$u_k^\top p = \frac{1-c_k}{\sqrt{k(k-1)}},$$

and hence

$$\frac{u_k^\top b_i}{p_i} = \begin{cases} -(1-c_k)/\sqrt{k(k-1)}, & k < i, \\ -(i-c_i)/\sqrt{i(i-1)}, & k = i, \\ c_k/\sqrt{k(k-1)}, & k > i. \end{cases} \tag{14}$$

Dividing their squares by c_k/n produces the three terms of (10). Since the exact conductances dominate half of those of L_0, the inverse-Laplacian variational principle yields $L^\dagger \preceq 2L_0^\dagger$ on the zero-sum subspace. This proves the first part of Lemma 4.

For the aggregate bound, split A_i into the three sums in (10). Jensen's inequality and a change in summation order bound the first and third contributions by one each. For example,

$$\sum_i p_i \sqrt{\sum_{k<i} \frac{(1-c_k)^2}{k(k-1)c_k}} \le \sqrt{\sum_{k\ge 2} \frac{1}{k(k-1)}} \le 1,$$

where $\sum_{i>k} p_i \le c_k$. For the middle term, let $S_i = \sum_{j\ge i} p_j$. Since $c_i \ge S_i$,

$$\begin{aligned}\sum_{i\ge 2} p_i \frac{i-c_i}{\sqrt{i(i-1)c_i}} &\le \sqrt{2}\sum_i \frac{p_i}{\sqrt{S_i}} \\ &\le 2\sqrt{2}.\end{aligned}$$

The last inequality telescopes using $(S_i - S_{i+1})/\sqrt{S_i} \le 2(\sqrt{S_i} - \sqrt{S_{i+1}})$. The supplement gives every projection and boundary case.

Corollary 5 (Profile-aware risk)**.** *With A_i as in* (10)*, every stationary trajectory satisfies*

$$\begin{aligned}\mathbb{E}d_{\mathrm{TV}}(\widehat{p}_T, p) &\le \sqrt{\frac{n}{T}}\,\Psi(p), \\ \Psi(p) &:= \sum_{i\in\mathrm{supp}(p)} p_i\sqrt{A_i} \le 2 + 2\sqrt{2}. \qquad (15)\end{aligned}$$

Proof of Theorem 2. By Lemmas 3 and 4,

$$\begin{aligned}\mathbb{E}d_{\mathrm{TV}}(\widehat{p}_T, p) &\le \frac{1}{2}\sum_i \sqrt{\mathrm{Var}(\widehat{p}_T(i))} \\ &\le \sqrt{\frac{n}{T}}\sum_i p_i\sqrt{A_i} \le (2+2\sqrt{2})\sqrt{\frac{n}{T}}.\end{aligned}$$

3.1 Examples and the slow mode

The profile functional in (15) reveals why a global gap bound is loose. If p is uniform on all n labels, then $c_k = 1$ and direct substitution gives

$$A_i = 1 - \frac{1}{n}, \qquad \Psi(p) = \sqrt{1 - \frac{1}{n}}.$$

The winner chain is a lazy complete-graph walk. Its TV occupation risk is therefore bounded by $\sqrt{(n-1)/T}$, at the ordinary empirical-distribution scale.

At the other extreme, take

$$p_1 = \frac{1}{2}, \qquad p_j = \frac{1}{2(n-1)}, \quad j \ge 2. \qquad (16)$$

The indicator $Y_t = \mathbf{1}\{X_t = 1\}$ is itself a two-state Markov chain. From either aggregate state it flips with probability exactly $1/n$. Its nonconstant eigenvalue is $1-2/n$, so estimating the heavy mass from one trajectory has variance of order n/T. This is a genuine slow mode. Nevertheless, there is only one such aggregate coordinate. The remaining $n-1$ light coordinates leave their individual states at constant rate. The sum of all coordinate standard deviations remains $O(\sqrt{n/T})$, exactly as Lemma 4 certifies.

For a nearly point-mass target, the slow aggregate mode becomes still slower, but its stationary variance shrinks in the same proportion. In capped-mass language, the local denominator c_k decreases only where the corresponding tail mass also decreases. The left and middle telescopes in the proof are the finite-dimensional expression of this compensation.

These examples also separate estimation from independent sampling. The uniform target behaves nearly as if the retained states were independent. The half-heavy target has only about T/n effective observations of its heavy-versus-light split. That split is precisely the source of the high-confidence lower bound in Section 5.

4 Burn-in and Confidence Amplification

The winner chain admits the pointwise minorization

$$P(i,\cdot) \ge \frac{1}{n-1}p(\cdot). \qquad (17)$$

This also holds from a zero-mass state. Hence, from every initial law μ,

$$d_{\mathrm{TV}}(\mu P^B, p) \le e^{-B/(n-1)}. \qquad (18)$$

Applying the same Markov kernel to two initial states cannot increase TV, so (18) also bounds the distance between the entire post-burn path and a stationary path.

With the B, T in Algorithm 1, Theorem 2 and (18) give

$$\mathbb{E}d_{\mathrm{TV}}(\widehat{p}_r, p) \le \varepsilon/24.$$

Markov's inequality shows that each candidate is within $\varepsilon/4$ with probability at least $5/6$. Hoeffding's inequality implies that at least $2R/3$ candidates are good except with probability $e^{-R/18} \le \delta$.

If G of R points in any metric space lie within radius r of p, where $G > R/2$, every geometric median q satisfies

$$d(q,p) \le \frac{2G}{2G-R}r. \qquad (19)$$

Indeed, compare its sum of distances with the sum at p and apply the triangle inequality separately to the good and bad points. With $G \geq 2R/3$ and $r = \varepsilon/4$, (19) proves the upper bound in Theorem 1.

Computation. The chain uses one oracle call and $O(1)$ additional work per transition if a uniform challenger can be sampled in constant time. The candidate histograms require $O(Rn)$ storage in a direct implementation. The geometric median in (19) is the linear program

$$\min_{q \in \Delta_n, z \geq 0} \sum_{r,i} z_{ri} \quad \text{subject to} \quad -z_{ri} \leq q_i - \widehat{p}_r(i) \leq z_{ri}.$$

Thus the statistical upper bound is also polynomial-time. Approximate minimization to $o(\varepsilon R)$ preserves the guarantee up to a constant change in accuracy.

Why ordinary confidence is additive. For T i.i.d. samples, the empirical TV risk is $O(\sqrt{n/T})$. Changing one sample changes TV loss by at most $1/T$, so bounded differences adds only $O(\sqrt{\log(1/\delta)/T})$ to this expectation (Canonne, 2020). This gives the sum $n + \log(1/\delta)$. Winner-chain states do not have this product concentration. The half-heavy example in (16) retains one transition for $\Theta(n)$ steps on average. The lower bound below shows that no different adaptive estimator can bypass the resulting confidence cost.

5 The Confidence Cost Is Necessary

We now show that no adaptive pair selection can attain the additive confidence dependence of ordinary sampling. For $a \in (0,1)$, define

$$p_a(1) = a, \qquad p_a(j) = \frac{1-a}{n-1}, \quad j \geq 2. \tag{20}$$

Take $a_- = 1/2 - 2\varepsilon$ and $a_+ = 1/2 + 2\varepsilon$. Then $d_{\mathrm{TV}}(p_{a_-}, p_{a_+}) = 4\varepsilon$.

A query not containing label 1 returns a fair draw under both hypotheses. For $\{1, j\}$, let the response be one when the light label wins. Its Bernoulli parameter is

$$r(a) = \frac{1-a}{1+(n-2)a}. \tag{21}$$

On $a \in [1/4, 3/4]$, one has $r(a) = \Theta(1/n)$ and

$$|r'(a)| = \frac{n-1}{(1+(n-2)a)^2} = O(1/n). \tag{22}$$

The elementary Bernoulli bound $D_{\mathrm{KL}}(x\|y) \leq (x-y)^2/[y(1-y)]$ therefore gives

$$D_{\mathrm{KL}}(\mathrm{Bern}(r(a_-))\| \,\mathrm{Bern}(r(a_+))) \leq C_1 \frac{\varepsilon^2}{n}. \tag{23}$$

For an adaptive learner, condition on each realized transcript before the next query. Every possible pair obeys (23), so the KL chain rule bounds the divergence between length-Q transcript laws by $C_1 Q\varepsilon^2/n$.

If an estimator is ε accurate with probability $1-\delta$ under both hypotheses, choosing the closer of p_{a_-} and p_{a_+} produces a test whose two error probabilities are at most δ. The Bretagnolle–Huber inequality then requires transcript KL at least $\log(1/(4\delta))$. Consequently,

$$Q \geq c_1 \frac{n \log(1/\delta)}{\varepsilon^2},$$

which proves the lower bound in Theorem 1.

6 Bounded Menus Interpolate the Obstruction

The confidence phenomenon is not unique to pairs. It weakens continuously when a query may expose more light labels at once. Let $q_{\leq k}(n, \varepsilon, \delta)$ denote minimax labeled TV complexity when the learner may condition on any nonempty menu of size at most k.

Proposition 6 (Bounded-menu confidence lower bound)**.** *For $2 \leq k \leq n$ and the parameter ranges of Theorem 1,*

$$q_{\leq k}(n, \varepsilon, \delta) = \Omega\left(\frac{n + (n/k)\log(1/\delta)}{\varepsilon^2}\right). \tag{24}$$

Proof. The $\Omega(n/\varepsilon^2)$ term already holds with unrestricted conditional access, since learning all labels is no easier than the standard multinomial packing used for ordinary samples (Canonne, 2020; Kleinberg et al., 2026).

For confidence, use the family (20). A menu excluding label 1 is uniform under both hypotheses. Suppose a menu contains label 1 and $s \leq k-1$ light labels. Conditional on a light outcome, its identity is uniform under both hypotheses, so all information is in the Bernoulli light probability

$$r_s(a) = \frac{s(1-a)}{(n-1)a + s(1-a)}. \tag{25}$$

Uniformly for $a \in [1/4, 3/4]$, one has $r_s(a) = \Theta(s/n)$ and

$$|r_s'(a)| = \frac{s(n-1)}{((n-1)a + s(1-a))^2} = O(s/n).$$

The Bernoulli KL is therefore $O(s\varepsilon^2/n)$, and hence at most $O(k\varepsilon^2/n)$, for every conditional menu and every transcript. The KL chain rule and Bretagnolle–Huber argument from Section 5 require $Q = \Omega((n/k)\log(1/\delta)/\varepsilon^2)$. Taking the maximum of the two lower bounds proves (24). ☐

For $k = 2$, Theorem 1 matches (24). For $k = n$, the lower bound becomes the ordinary additive rate. This identifies a candidate interpolation law, but we do not claim its upper bound for intermediate k. A direct k-winner chain has stationary law p and a stronger minorization, yet its coordinate Green function is no longer the minimum-mass graph in Lemma 4. Determining whether its occupation profile attains (24) is a separate open problem.

7 Related Work and Scope

Restricted conditional learning. Conditional sampling was introduced for distribution testing (Canonne et al., 2015; Chakraborty et al., 2016). The PCOND restriction is powerful for many label-invariant testing problems, but our target is the entire labeled distribution. Kleinberg et al. (2026) characterize which fixed menu families permit pointwise and uniform labeled learning. Their global and local winner chains motivate ours. Their global learner restarts after every terminal sample, while their hierarchical learner keeps local trajectories to estimate tree splits. Our result instead controls the full empirical law of one global pairwise trajectory through its complete mass profile. It both removes the dimension logarithms and supplies the matching confidence lower bound.

Comparison estimation and sampling. Classical BTL work estimates rankings, log qualities, or normalized weights under passive comparison graphs. Sharp bounds depend on graph Laplacians, effective resistance, and often bounded dynamic range (Shah et al., 2015; Negahban et al., 2017; Agarwal et al., 2018; Hendrickx et al., 2019, 2020). Our active estimator permits all simplex boundary points and is analyzed in labeled TV. Fotakis et al. (2022) construct exact samples from exogenously drawn pair comparisons using coupling from the past. Their distribution-free method first learns every weight to relative accuracy and solves a different exact-sampling objective.

Biased sampling and Markov concentration. Conditional menus are a finite selection-bias model. Classical work proves identifiability and asymptotic efficiency for fixed biased-sampling designs (Vardi, 1985; Gill et al., 1988). Generic reversible-chain concentration scales with a worst-case spectral gap (Jiang et al., 2018). That route loses a factor n here. The capped mass calculation is coordinate-specific and nonasymptotic.

Limits and open directions. Theorem 1 concerns the family of all pairs. It does not close the polylogarithmic gaps for every hierarchically comparable menu family. Proposition 6 gives a broader necessary interpolation, but whether it is sufficient for every menu cap remains open. The present algorithm also assumes that every pair is available. Graph-restricted active designs may require additional connectivity and dynamic-range parameters.

8 Conclusion

Pairwise conditional samples retain enough aggregate information to learn a labeled distribution at the ordinary n/ε^2 rate at constant confidence. They do not retain the ordinary model's cheap confidence amplification. Reusing a single winner-chain trajectory attains the optimal constant-confidence rate, while independent trajectories and a metric median match the unavoidable multiplicative confidence cost. The proof explains why a linear relaxation time does not imply a linear loss in estimation risk: the slow modes and the TV sensitivity are coupled through the target's capped mass profile.

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